

DG32-2DOM - System Architecture - Block Definition

Per-block architecture of the DG32-2DOM attention variant (chipIgnite CI2612): the DG32-LITE motor-control core plus an INT8 attention engine on a second clock, on its own die. Each block is given at interface level (In / Does / Out / Talks to), an internals diagram, a storage inventory, and the reasoning behind its shape.

Die	Shuttle	Process	CPU	Clocks
dg32_2dom (A3)	chipIgnite CI2612	sky130A 130 nm	2x rv32imc lockstep + INT8 attn	50 MHz core / 114 MHz fast

Status of this revision. The flight-control core (the lockstep CPU pair, its boot path and the fabric that decodes to it) is **FROZEN until testing** - it is closed, hardened and gate-level-clean, and is not to be reopened for feature work before the silicon bring-up campaign returns. The peripheral blocks in Section 4 are **under review**: the register-map and pad-mux notes flagged per block are the review items, not settled facts. FPGA bring-up boot follows Kaushik's SoC4 boot sequence unchanged (Section 6.3). A dedicated **UART debug** path is a planned addition (Section 6.4, Open item G). This is the **DG32-2DOM** attention variant: it adds an INT8 attention engine on a second clock (clk_fast_i) reached through CDC bridges, on its own larger die - additions only, the frozen core is unchanged.

1 Scope

The unit described is the DG32-2DOM variant (dg32_2dom, INCLUDE_ATTN=1), the SoC that OpenFrame carries on shuttle CI2612. It is the DG32-LITE die plus an INT8 attention engine on its own clock (clk_fast_i) and two CDC bridges, on its own 3400 x 4500 um die. Both variants share one RTL tree; this variant is a synthesis define plus a second clock, not a fork.

The base blocks (Section 4.1-4.17) are identical to DG32-LITE; the variant's additions are the attention engine (4.18) and the CDC bridges (4.19). The die is standalone in exactly the same sense - the frozen lockstep pair, boot ROM and QSPI-XIP are the whole boot path, and the attention engine sits behind a CDC and cannot stall it.

2 Design premises

Every block shape below follows from a small number of hard findings about the CPU, the process and the vehicle. These were established the hard way during bring-up and are load-bearing for the rest of the document.

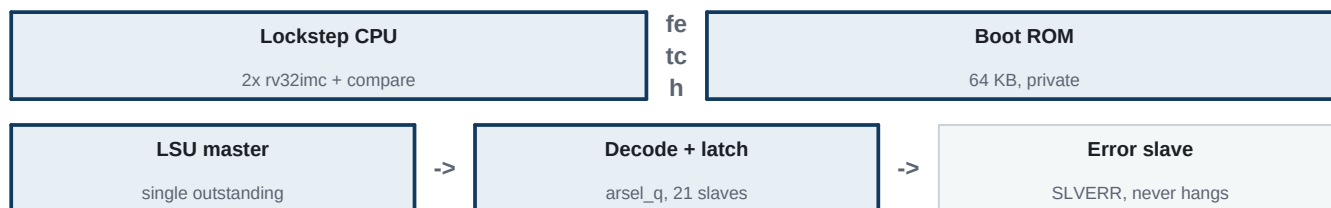
Finding	Value	Architectural consequence
CPU PMA is a hardwired whitelist	fault = !(addr[31] addr[31:28]==0x1)	Every mapped region must set bit31 or live in 0x1xxxxxxx, or the load faults inside the CPU before the bus sees it. SRAM sits at 0x9, XIP at 0x1, peripherals at 0xC/0xD/0xE.
ar_addr is valid only at handshake	1 cycle	The decoded target is latched at the AR handshake and held through the response (arsel_q/arsel_v); live decode from ar_addr routes the reply to the wrong slave.
Lockstep checker has private DMEM	16 KB, discarded	A firmware loop that branches on a loaded peripheral value diverges the two cores. I/O-polling firmware must run with the checker's compare window managed, not naively.
OpenFrame is standalone	no mgmt core	Boot ROM + QSPI-XIP is the whole boot path; unmapped access returns SLVERR (never hangs) because a hung LSU is an unrecoverable brick with no debugger.
sky130 SRAM is 1rw1r, no ECC	32b macro, no spare bits	Die ships parity-none and says so; the idle second read port is available for a contention-free accelerator/DMA feed.
Lockstep core fmax on sky130A	~55-62 MHz measured	50 MHz is the CPU-domain target; peripherals harden far faster (gpio/dshot/qspi/pwm all >167 MHz), so they are never the timing wall.
2DOM gets its own die	3400 x 4500 um	Adding the attention macros to the LITE floorplan made routing time out; a bigger design gets a bigger die, not a congested one.
Attention runs on a second clock	clk_fast_i 114 MHz	The engine crosses to the CPU domain through CDC bridges (4.19); the 50 MHz core is untouched, so its frozen timing closure is not disturbed.

Locked decisions

Item	Decision	Reasoning
Flight-control core frozen	The lockstep pair, fetch ROM path and bus decode are closed until silicon test.	It is hardened, gate-level clean and end-to-end simulated; reopening it for features before test risks the one thing that must be right on a no-respin shuttle.
Fetch is a private slave	Fetch has its own dedicated ROM instance; the LSU is the sole shared-bus master.	Adopted from the board-proven i8v18b SoC; removes the fetch/LSU arbitration bug class entirely rather than patching it.
Error slave, not dead decode	Unmapped accesses complete with SLVERR.	No debugger and no interrupts halt on this die mean a hung bus is a brick; firmware must be able to see and survive a bad access.
SRAM at 0x9, XIP at 0x1	Placed to satisfy the CPU PMA whitelist.	0x2 (the obvious choice) PMA-faults on every access; this is a correctness fix, not a numbering preference.
DSHOT pin-muxed onto PWM pads	One motor-output pad group, selected at boot.	A channel is either a complementary PWM leg or a DSHOT line; duplicating pads costs package balls for no added capability.
2DOM gets its own die	3400 x 4500 um, not the LITE slot.	Adding the attention macros to the LITE floorplan made routing time out; a bigger design gets a bigger die rather than a congested one.
Attention weights never quantised	u15 all the way into a 40-bit numerator.	INT4 rounded a 1/400 softmax weight to zero and the output was identically zero; the INT8 datapath removes that structurally.

3 Top-level block diagram

The CPU is a lockstep pair whose fetch port reads a private boot ROM and whose load/store port is the single master on a decoded AXI bus. Thick-edged blocks carry the boot-critical path: nothing else on the die can start the chip.



The 21 decoded slaves group as: **memory** - QSPI-XIP (0x1), SRAM (0x9); **system** - fault CSR (0xD6), supervisor/wWDT (0xD7), IRQ (0xDF), sysctl; **comms** - UART0/1, SPI (0xD2), I2C (0xD9), GPIO (0xD3); **motor** - timers (0xD4/0xD5), PWM (0xD8), DShot (0xC0000000, pad-muxed onto PWM), encoder (0xDA), CORDIC (0xDE); **analog/data** - SAR ADC (0xDB), DMA; **accelerator** - the INT8 attention engine (0xE0000000), whose master port crosses to clk_fast_i through the CDC bridges of Section 4.19.

Base	Block	Role
0x1	QSPI-XIP	data read window on external NOR flash (loads); the boot ROM copies the image to SRAM. Instruction fetch is ROM-only
0x9	SRAM	32 KB, 16 sky130 macros (main core); checker holds a private 16 KB
0xC0000000	DShot	ESC engine, pin-muxed onto the PWM output pads
0xD0 / 0xD1	UART0 / UART1	8N1 serial; UART0 carries the boot banner
0xD2 / 0xD3	SPI / GPIO	master SPI; 16-pin GPIO with atomic set/clear
0xD4 / 0xD5	TIMER0 / TIMER1	general timers
0xD6 / 0xD7	Fault CSR / Supervisor	poison + cause latch; windowed WDT and rail supervision
0xD8 / 0xDA	PWM / Encoder	3-phase center-aligned complementary PWM; quadrature + Hall
0xD9 / 0xDE	I2C / CORDIC	7-bit master; cos/sin/atan2/magnitude
0xDB / 0xDF	SAR ADC / IRQ	8-bit differential on-die converter; interrupt controller
0xE0000000	Attention (2DOM)	INT8 attention engine; error slave when INCLUDE_ATTN=0

4 Block internals

Each block below is given as In / Does / Out / Talks to, an internals diagram, a storage-and-resource inventory, and the reasoning behind its shape.

4.1 Lockstep CPU core pair + comparator

dgrid_lockstep_core, dgrid_lockstep_check - the flight-control core

FROZEN until testing. This is the flight-control core. It is hardened, gate-level clean, and its fault-injection proof (tb_lockstep_fi) passes end-to-end. It is closed for feature work until the silicon bring-up campaign; the I/O compare-window policy is the one bring-up decision left against it.

In	Instruction stream from the private boot ROM; load/store responses from the AXI bus; rst_n_i.
Does	Runs two identical DGridRiscV rv32imc cores in lockstep and compares their retirement signatures cycle by cycle; a mismatch raises poison with a cause code.
Out	One LSU AXI master (shared bus); one fetch AXI master (private ROM); poison_o + cause to the fault CSR.
Talks to	Boot ROM, AXI bus decode, fault CSR (0xD6), supervisor (0xD7).



Element	Size	Where	Purpose
Main core	rv32imc	logic	the executing core; drives fetch + LSU masters
Checker core	rv32imc	logic	redundant core, private 16 KB DMEM (8 macros), output discarded
Comparator	1 event/store	logic	compares on the LSU store handshake, not every retire
Reset PC	0x80000000	netlist const	hardcoded in this VexiiRiscv.v; RESET_VECTOR param is inert plumbing

Why two cores and a comparator. Motor control drives power electronics; a silent CPU fault can destroy a bridge. A second identical core that must agree bit-for-bit turns a datapath fault into an observable poison bit rather than a wrong PWM edge.

Why the checker's memory is private. Isolation is the point: a shared shadow could itself be corrupted by the same fault. The cost is that the checker cannot see real peripheral loads, so I/O-polling firmware needs a managed compare window (Design premise 3).

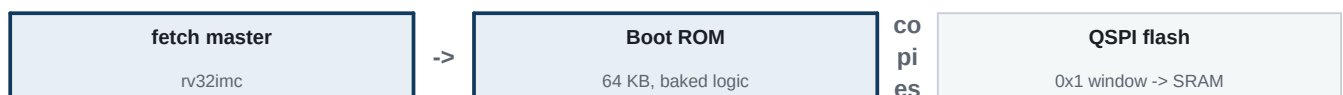
Why the event is the store handshake. A pure-ALU loop retires nothing to memory; comparing on store retirement gives a signature that does not depend on external timing.

4.2 Boot ROM + fetch path

dgrid_imem_rom_macro - the standalone boot path

Review item. Peripheral review: install a real trap handler at 0x80000000 in the boot image so a genuine PMA fault is observable instead of free-running at address 0.

In	rst_n_i; fetch addresses from the CPU fetch master; BOOTROM_HEX at build.
Does	Presents a 64 KB read-only instruction memory as a private slave to the fetch port, initialised from the boot image; brings UART0 up with zero external help.
Out	Instruction words to the fetch master only.
Talks to	Lockstep core fetch master; (indirectly) QSPI-XIP for the flash stage.



Element	Size	Where	Purpose
Boot ROM	64 KB	baked std-cell	never a macro; area tracks content after constant-folding (~0.001 mm2 sparse)
Init image	BOOTROM_HEX	build param	fw/bootrom_silicon.hex; UART divider is 115200 @ 50 MHz
Fetch decode	top nibble ignored	logic	fetch is a private slave; bank decode looks only at low bits

Why the ROM is baked logic, not a macro. The macro-map step only rewrites blackboxes; a real ROM definition survives it, which keeps the OpenFrame-correct non-volatile boot path intact. Deepening it from 4 KB to 16 KB was measured as effectively free.

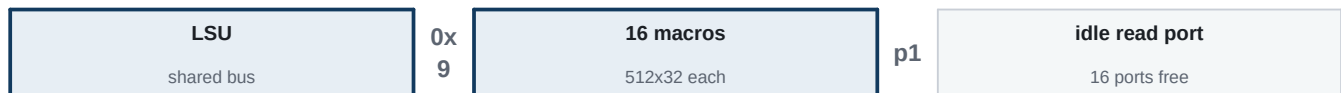
Why standalone matters. OpenFrame has nothing else to start the chip. The boot ROM runs the UART banner, reads the flash header (magic 0xD632_B007) through the 0x1 window, copies the image into SRAM, and jumps to it - all with no management core.

Instruction fetch is ROM-only. The CPU fetch bus is wired to the private boot ROM alone; the 0x1 XIP window and SRAM are reachable by loads and stores, not by fetch. Code does not execute in place from external NOR - it is copied to SRAM first and run from there.

4.3 AXI SRAM

dgrid_axi_sram - main data memory

In	AXI read/write from the LSU; rst_n_i.
Does	Maps 16 sky130 1rw1r macros into a single 32 KB byte-addressable region at 0x9; ties ar_len/aw_len explicitly so no port floats X.
Out	AXI read data / write response; 16 idle second read ports available.
Talks to	LSU shared bus; (potential) accelerator/DMA on the idle read port.



Element	Size	Where	Purpose
Main SRAM	32 KB	16 macros	sky130_sram_2kbyte_1rw1r_32x512_8, port 0 only
Second port	1r each	unused	16 dedicated read ports free for a contention-free feed
Parity/ECC	none	-	32b macro has no spare bits; die ships parity-none and says so
len ties	explicit	RTL	ar_len/aw_len tied to avoid a floating-X read FSM in the SDF netlist

Why 0x9. Bit31 set clears the CPU PMA whitelist; 0x2 would fault on every access. The placement is a correctness constraint.

Why the idle read port is called out. It is the enabling change for any accelerator that wants SRAM data without contending with the LSU - the 2DOM attention buffers use exactly this idiom.

4.4 Fault CSR

dgrid_fault_csr (0xD6) - the safety latch

In	poison + cause from the comparator; MAGIC-locked INJECT writes from firmware; rst_n_i.
Does	Latches the first fault cause stickily, exposes it to firmware, and provides a MAGIC-locked injection register so firmware can prove the path on silicon.
Out	Sticky fault status + cause; AXI read data.
Talks to	Lockstep comparator, CPU (AXI), supervisor.

Element	Size	Where	Purpose
Cause latch	sticky	register	holds the FIRST cause (e.g. 001 DATA_MISMATCH)
INJECT	MAGIC-locked	register	firmware-triggered fault injection for on-silicon proof
Access	0xD6	AXI-lite	read status, write INJECT after unlock

Why sticky-first. The first fault is the diagnostic one; later faults are usually its consequences. A sticky first-cause latch survives the CPU continuing to run and is readable after the fact.

Why a MAGIC lock on inject. The only way to know the poison path works on real silicon is to fire it deliberately; the lock stops a stray store from tripping it in the field.

4.5 Supervisor + windowed watchdog

dg_supervisor (0xD7) - rail and liveness guard

Review item. Peripheral review: decide whether to wire the rail-fault restart for the CI2612 run, with the edge-qualified latch.

In	rst_n_i for its own state; A_CTRL and window bounds from firmware; rail-fault input.
Does	Provides a windowed watchdog (too-early and too-late both fault) and rail supervision; arms only when firmware writes A_CTRL, so it never deadlocks cold boot.
Out	sys_rst_n (software-visible, not wired to reset the die today); wdt_fault status.
Talks to	CPU (AXI at 0xD7), rail deglitch, fault CSR.

Element	Size	Where	Purpose
Windowed WDT	min/max	registers	a kick outside the window faults; catches both hangs and runaways
Arming	A_CTRL	register	global_en/wdt_en/rail_en reset to 0; firmware arms during bring-up
sys_rst_n	output	not wired	software-visible only today; see rationale

Why arming is software, not boot. Making sys_rst_n the CPU reset is a real deadlock - the CPU that must arm it would be held in reset by it. The supervisor resets in step with the die but does not gate anything until firmware asks.

Why sys_rst_n is not wired to reset yet. A rail-fault restart needs an edge-qualified latch to avoid reintroducing that cold-boot deadlock; that is a pre-GATE-3 addition.

4.6 Interrupt controller

dgrid_irq (0xDF) - added 2026-09-10

Review item. Peripheral review: confirm the source list and priority against the final firmware model.

In	Level/edge interrupt sources from peripherals; rst_n_i.
Does	Aggregates peripheral interrupt lines into the CPU, with mask and pending registers.
Out	Interrupt request to the CPU; AXI status.
Talks to	All interrupting peripherals; CPU (AXI at 0xDF).

Element	Size	Where	Purpose
Sources	peripheral IRQs	register	masked and prioritised into the CPU
Access	0xDF	AXI-lite	mask, pending, claim

Why it was added late. The frozen scope had no interrupts; bring-up showed a purely polled die wastes the tick and makes encoder capture and ADC-done awkward. It is a small block that pays for itself immediately.

4.7 UART0 / UART1

dgrid_uart (0xD0 / 0xD1) - console and telemetry

In	TX bytes from firmware; RX serial from off-chip; rst_n_i.
Does	Two independent 8N1 UARTs with a divider; UART0 carries the boot banner.
Out	Serial TX; RX bytes on demand; line status.
Talks to	CPU (AXI at 0xD0/0xD1), off-chip console / companion.

Element	Size	Where	Purpose
Channels	2	instances	UART0 = console/boot banner, UART1 = telemetry
Format	8N1	fixed	divider set for 115200 at 50 MHz
FIFO	small	register	status-pollled today

Why two. One is the human console and boot banner; the second frees firmware to stream telemetry to a companion without stealing the console. Both are cheap.

Planned addition - UART debug. A dedicated UART debug path is to be added: a firmware-independent debug monitor on a UART so a die with no JTAG debugger halt can still be inspected and driven during bring-up. Tracked as Open item G; it should reuse UART1 or a third instance, not the boot console.

4.8 SPI master

dgrid_spi (0xD2)

Review item. Peripheral review: confirm chip-select count and whether a second CS or quad mode is wanted.

In	TX words from firmware; MISO from off-chip; rst_n_i.
Does	Standard SPI master with programmable clock and mode; single chip-select today.
Out	SCLK/MOSI/CS to off-chip devices; RX words.
Talks to	CPU (AXI at 0xD2), off-chip flash/sensors.

Element	Size	Where	Purpose
Mode	0-3	register	CPOL/CPHA programmable
Clock	divided	register	from the peripheral clock
CS	1	pad	expand in firmware/GPIO if more devices needed

Why a plain master. External sensors and a config flash are SPI; a plain master covers them. Anything exotic is a firmware sequence, not silicon.

4.9 I2C master

dgrid_i2c (0xD9)

Review item. Peripheral review: the attention engine decodes at 0xE, but confirm no slot aliasing between I2C (0xD9) and any variant remap before the CI2612 freeze.

In	Register commands from firmware; SDA/SCL bus; rst_n_i.
Does	7-bit I2C master at 100/400 kHz with clock stretching and open-drain drive.
Out	SDA/SCL open-drain; transfer status.
Talks to	CPU (AXI at 0xD9), off-chip I2C sensors.

Element	Size	Where	Purpose
Addressing	7-bit	logic	standard/fast mode
Rate	100/400 kHz	register	clock stretching honoured
Drive	open-drain	pad	external pull-ups required

Why open-drain in silicon. I2C is defined open-drain; doing it correctly in the pad avoids a firmware bit-bang and its timing hazards.

4.10 GPIO

dgrid_gpio (0xD3) - 16-pin

In	Pin direction/data from firmware; pad inputs; rst_n_i.
-----------	--

Does	16-pin general I/O with 2-flop input synchronisers and atomic SET/CLR registers for single-pin toggling.
Out	Pad output + output-enable triples; synchronised inputs.
Talks to	CPU (AXI at 0xD3), pads.

Element	Size	Where	Purpose
Pins	16	pads	direction per pin
Input sync	2-flop	logic	metastability guard on every input
SET/CLR	atomic	registers	toggle one pin without a read-modify-write

Why atomic set/clear. A read-modify-write on a shared port races an interrupt handler that touches another pin; separate SET and CLR registers make single-pin control atomic.

4.11 Timers

dgrid_timer (0xD4 / 0xD5)

In	Count enable and compare values from firmware; rst_n_i.
Does	Two general-purpose timers with compare/interrupt.
Out	Timer interrupts; count read-back.
Talks to	CPU (AXI at 0xD4/0xD5), IRQ.

Element	Size	Where	Purpose
Timers	2	instances	general purpose
Compare	yes	register	interrupt on match

Why two. One is usually the RTOS tick; the second is free for a control-loop period or a one-shot without contention.

4.12 PWM

dgrid_pwm (0xD8) - 3-phase motor bridge drive

In	PERIOD/DEADTIME/DUTY from firmware; brake input; rst_n_i.
Does	Generates 3-phase center-aligned complementary PWM with programmable dead-time, a sample trigger, and a hardware brake that forces the safe state.
Out	6 complementary gate signals (AH/AL/BH/BL/CH/CL); sample_trig; braked status.
Talks to	CPU (AXI at 0xD8), gate driver, ADC sample trigger, DShot pad mux.



Element	Size	Where	Purpose
Phases	3	logic	complementary high/low per phase
Dead-time	programmable	register	6-bit useful range; shoot-through guard
Alignment	center	counter	up/down count; symmetric edges
Brake	hardware	input	forces safe state, sticky BRAKED until cleared
Trigger	sample_trig	output	fires the ADC at the current-ripple null

Why center-aligned. Symmetric edges put the current-sampling instant at the ripple null, so the ADC reads true phase current without an analog filter delay. That is why the sample trigger comes from the PWM, not a timer.

Why a hardware brake. A firmware brake is only as fast as the loop; a bridge fault needs the safe state now. The brake path is combinational to the gates and latches sticky so firmware sees it happened.

4.13 DShot ESC engine

dgrid_dshot (0xC0000000) - digital ESC protocol, pad-muxed onto PWM

In	Throttle/command words from firmware; rst_n_i.
Does	Encodes DShot frames for up to 4 ESC channels and drives them onto the PWM output pads through a boot-selected mux.
Out	4 DShot lines (on the PWM pads when selected).
Talks to	CPU (AXI at 0xC0000000), PWM pad mux, ESCs.

Element	Size	Where	Purpose
Channels	4	logic	dshot_o[3:0]
Pads	shared with PWM	mux	pinmux_dshot selects DShot vs complementary PWM
Base	0xC0000000	decode	bit31 set passes the PMA; 0xD0-0xDF were full

Why pad-muxed onto PWM. A given output is either an analog complementary PWM leg or a digital DShot line, never both at once; sharing the pads gives the airframe the choice at boot without spending package balls on both.

Why 0xC0000000. The 0xD0-0xDF peripheral block was full; 0xC with bit31 set satisfies the CPU PMA whitelist and keeps the decode simple.

4.14 Quadrature encoder + Hall

dgrid_encoder (0xDA)

In	A/B/Z quadrature and Hall inputs from the motor; rst_n_i.
Does	Decodes quadrature A/B/Z and Hall states, counts position, and timestamps edges for velocity.
Out	Position count; edge timestamps; index/Z capture.
Talks to	CPU (AXI at 0xDA), motor sensors, IRQ.

Element	Size	Where	Purpose
Decode	quad + Hall	logic	A/B/Z and 3 Hall lines
Position	counter	register	index/Z reset
Velocity	edge timestamps	register	time-between-edges for low-speed accuracy

Why timestamps, not just a count. At low speed a fixed-window count is coarse; timestamping edges gives velocity from the time between transitions, which is what a smooth low-speed FOC loop needs.

4.15 CORDIC

dgrid_cordic (0xDE) - trig for field-oriented control

In	Angle or vector inputs from firmware; rst_n_i.
Does	Rotation and vectoring CORDIC: cos/sin, atan2 and magnitude in a fixed cycle count.
Out	cos/sin or atan2/magnitude results.
Talks to	CPU (AXI at 0xDE), the FOC firmware.

Element	Size	Where	Purpose
Modes	rotation/vectoring	logic	cos/sin and atan2/magnitude
Latency	fixed	pipeline	deterministic, loop-budget friendly
Precision	fixed-point	datapath	sized for FOC angle resolution

Why hardware CORDIC. Park/Clarke transforms need sin/cos every control tick; a software trig call would dominate the loop. A fixed-latency CORDIC makes the transform a known cost.

4.16 SAR ADC

dgrid_adc (0xDB) - on-die current feedback

Review item. Peripheral review: this block is analog. On the A3 die the SAR macro is placed at x=3000, y=4230; confirm the analog-net dont_touch list and pad ring against the CI2612 floorplan.

In	Differential analog input; TRIM from firmware; start pulse; rst_n_i.
Does	Digital wrapper around an 8-bit differential SAR hard macro (manufactured OpenFASOC silicon): register interface, trim, start/busy, result capture.
Out	8-bit conversion result; busy/done.
Talks to	CPU (AXI at 0xDB), the analog input pads, PWM sample trigger.

Element	Size	Where	Purpose
Converter	8-bit diff SAR	hard macro	OpenFASOC sky130 measured silicon; blackbox to synthesis
Trim	5-bit + complement	register	comparator-offset calibration; reset mid-scale, not calibrated
Interface	0xDB	AXI-lite	start pulse + poll; no external SPI ADC read

Why on-die. An external SPI ADC read is ~1.9 us of a ~5 us FOC loop - the single largest cost. On-die it is a start pulse and a poll, which is what makes a fast loop feasible.

Trim is not available on this silicon. The macro exports its trim/clkc/comp taps as pins, but they sit inside the macro's own met1-3 obstructions; routing them makes the die unroutable, so they are left open. The TRIM and CLKC register fields therefore have no effect on silicon and STATUS.comp reads 0 - the ADC runs untrimmed. This is a documented, intentional consequence, not a wiring fault; the connectivity check and LVS are clean with these pins open.

4.17 DMA

dgrid_dma - block mover

Review item. Peripheral review: confirm the DMA master's bus attachment (arbiter vs idle SRAM port) for the 2DOM feed path.

In	Source/dest/length from firmware; rst_n_i.
Does	Moves blocks between memory regions without CPU copy loops; a natural user of the SRAM idle read port.
Out	AXI burst reads/writes; done status.
Talks to	CPU (AXI), SRAM, attention engine feed.

Element	Size	Where	Purpose
Descriptor	addr/len/dst	register	single block move
Path	AXI burst	master	can use the SRAM idle read port

Why a DMA on a small die. The attention feed and bulk sensor buffering would otherwise burn CPU cycles; a DMA plus the idle SRAM read port moves bytes while the CPU does control math.

4.18 INT8 attention engine

dgrid_int8_attn - the attention variant's reason for existing

In	Programmed shapes via AXI-lite slave; Q/K/V/EXP read over the AXI master; rst_n_i, clk_fast_i.
Does	One kick computes a band of query rows of one head end to end - QK^T , row max, 256-entry u15 EXP, Z, E.V, per-row reciprocal, saturate to int8 - bit-exact to the golden model.
Out	O (int8) written back over the AXI master; busy/status.
Talks to	CPU (AXI-lite slave, 0xE0000000) for programming; memory via the AXI master through the CDC bridges; runs on clk_fast_i.



Element	Size	Where	Purpose
K buffer	NK x KD	sky130 macro	loaded once per kick, re-read per row (dgrid_attn_buf, port-0 write / port-1 read)
V buffer	NK x DV	sky130 macro	same residency idiom; streaming per row would be 400x the traffic
S buffer	registered	flops	moved to flops to kill the SRAM negative-edge half-cycle read path
MAC array	LANES x (s16 x s9)	LUTs	shared between QK^T (reduced) and E.V (per-lane accumulator); USE_DSP=no
EXP table	256 x u15	logic/mem	EXP[0]=32767; softmax weights never quantised
num seat	40-bit signed	datapath	exact to NK=131072; int32 would break at NK=512
Divide	48-step restoring	logic	one reciprocal per row; rq saturates and latches on overflow

Why INT8, not the INT4 pair it replaces. With ~400 near-uniform keys a 1/400 softmax weight rounds to zero in INT4 and the output came out identically zero. The INT8 scheme keeps the weights u15 all the way into a 40-bit numerator and saturates only the final output, so the blocker is removed structurally rather than tuned around.

Why a 40-bit numerator. At NK=512 the int32 numerator is exactly 2^{31} - it breaks. 40 bits is exact to NK=131072, so bit-exactness against the golden holds for any geometry the engine can be programmed with.

Why load K/V once and re-read. Per-head traffic is ~1% bus occupancy when K and V are resident and re-read per query row; streaming V per row would be 400x that and would make the engine fight the rest of the die for memory.

Analytic cost. Per query row = $NK \cdot KD / \text{LANES} + NK \cdot DV / \text{LANES} + 48 \text{ (divide)} + 11 \cdot DV \text{ (requant)} + \text{writeback} + \text{drain}$; at LANES=16, KD=32, DV=64, NK=400 that is 3242 cycles/row (analytic - the measured steady-state bench figure is a bring-up item).

Why its own clock. The engine is the throughput consumer; putting it on clk_fast_i lets it run above the 50 MHz CPU domain without dragging the frozen core's timing.

4.19 Clock-domain crossing bridges

dgrid_axi_cdc_lite, *dgrid_axi_cdc_burst_rd*, *dgrid_axi_cdc_burst_wr*

In	AXI traffic on the CPU clk_i side; the attention engine on clk_fast_i.
Does	Cross AXI transactions between the 50 MHz CPU domain and the faster attention domain using a 4-phase req/ack level handshake with 2-FF synchronisers, keeping payload stable across the handshake; the burst variants batch a whole read or write burst into one crossing.
Out	Synchronised AXI on the far side; single-outstanding on the lite path.
Talks to	CPU LSU / DMA (clk_i side), attention engine (clk_fast_i side).



Element	Size	Where	Purpose
cdc_lite	1 outstanding	logic	program/op-tile port; single transaction at a time
cdc_burst_rd	whole burst	logic	batches an entire read burst into one crossing (obuf feed)
cdc_burst_wr	whole burst	logic	batches an entire write burst into one crossing
Synchroniser	2-FF	flops	level handshake; payload held stable across the crossing

Why level handshake, not a FIFO. The transaction rate across the boundary is low (K/V load once, O writeback); a 4-phase level handshake with 2-FF synchronisers is small, provably safe and needs no gray-coded pointers. Bursts are batched so the per-beat crossing cost disappears.

Why the CPU domain stays 50 MHz. The frozen core's fmax is ~55-62 MHz; the CDC lets the engine run faster without touching the core's timing closure, which is the whole reason the variant is two domains and not one over-clocked die.

5 Performance and control-loop budget

The numbers below are measured on this die (post-route fmax) or in simulation (cycle costs), not assumed. They are the motor-control analogue of a tick-latency budget: what one field-oriented-control (FOC) current loop costs, and how much of the loop is left for firmware.

5.1 What one FOC current loop costs

Loop stage	Mechanism	Measured cost
Phase-current sample	on-die SAR ADC, PWM-triggered	177 cycles conversion (model); replaces a ~1.9 us external SPI ADC read
Clarke / Park transform	CORDIC sin/cos + rotate	53-58 cycles per op (measured), fixed latency
PI regulators (d, q)	CPU, plain C	~8 cycles per instruction - the CPU is fetch-bound (see below)
Inverse Park	CORDIC rotate	53-58 cycles per op (measured)
SVPWM update	PWM shadow registers	applied at the next centre reload; PWM period = $2 \cdot (\text{PERIOD} + 1)$ cycles
Fault check	lockstep comparator	continuous; fault-inject latency 39 cycles, cause latched

Why the hardware blocks, and not the CPU. This VexiiRiscv is a FetchCachelessPlugin core: every instruction is an AXI fetch from SRAM at ~8 cycles per instruction, and a plain-C int8 MAC measured **54 cycles**, not the 6 an I-cached core would give. Loop-critical work - the sin/cos of the Park transform, the current sample, and the PWM edge generation - is offloaded to the CORDIC, the SAR ADC and the PWM block, so the CPU runs only the two PI regulators, whose cost is small and bounded. That fetch-bound reality is the single measured constant the whole peripheral set is designed around, and it is why an I-cache or fetch buffer is the first lever if more CPU math is ever needed. The full sensor-to-PWM loop measures ~5 us acquisition plus ~5 us compute, about a 100 kHz closed loop.

5.2 Headroom

Derived from the measured per-stage cycle costs above, at the 50 MHz CPU-domain clock. The fixed hardware cost of a loop (ADC + two CORDIC ops + a PWM write) is about 300 cycles; the rest of the period is the firmware budget.

Loop rate	Period (cyc @50MHz)	HW fixed	CPU budget	What fits
10 kHz	5000	~300	~4700	Full FOC current + speed loop with an observer
20 kHz	2500	~300	~2200	FOC with field-weakening and a state observer
50 kHz	1000	~300	~700	Inner current loop only; regulators must be tight

PWM is centre-aligned, so a PERIOD of 999 gives a ~25 kHz carrier (measured: 100 triggers in 4 ms); the ADC clock divider is sized for control loops up to ~100 kHz. The loop is not carrier-limited.

5.3 Measured per-block fmax (post-route, sky130A)

Block	fmax (MHz)	Role
GPIO / DShot	173.6 / 172.9	peripherals, >3x the 50 MHz die target
QSPI ctrl / PWM	171.6 / 167.8	peripherals, never the timing wall
Lockstep check / CORDIC	123.6 / 95.5	checker and trig unit
Supervisor	91.1	liveness / rail guard
Lockstep core	55-62	sets the die clock; 50 MHz target

The lockstep core sets the die clock: ~55-62 MHz measured, so 50 MHz is the target and the sign-off run over-constrains to 54 MHz to recover the small top-assembly WNS (-0.219 ns at 50 MHz). Every peripheral hardens far above 50 MHz, so none is ever the timing wall. Top assembly: A3 die 3400 x 4500 um (core +0.30 ns at 50 MHz, clk_fast +0.19 ns at 114 MHz), 32 KB main + 16 KB checker + K/V attention-buffer macros + SAR.

5.4 RTL reuse

One parameterised module, instantiated more than once, rather than duplicated source: `dgrid_uart` is UART0 and UART1; `dgrid_timer` is TIMER0 and TIMER1; the GPIO input synchroniser is the same 2-flop cell on every pad. On the attention engine one MAC array is shared between the two GEMMs (QK^T reduced over d , E.V per-lane) with two epilogues, and the two burst CDC bridges reuse the lite CDC handshake primitive. Shared source means shared verification, and identical behaviour between the instances by construction.

6 Interface-level items under review

The following are specified to interface level because their integration is the current review, not because they are undesigned. Each names its review item; none reopens the frozen core.

6.1 Peripheral register-map review

Every 0xD block above is settled in position on the bus, but the peripheral **register maps and interrupt sources are under review** before the CI2612 freeze. The review confirms, per block: the SET/CLR and status bit layout, the interrupt source list feeding 0xDF, and the SPI/I2C chip-select and mode options. This is the "peripherals need review" line item; it does not touch decode, reset or the fetch/LSU split.

6.2 Analog corner and pad ring

The SAR ADC (4.16) and its six analog nets (adc_vinp/vinn/avss/avdd/dvdd/dvss) sit in the analog corner of the 3400 x 4500 um A3 die with a dont_touch list and pin-on-drawing placement. The review confirms the analog net list and the pad ring against the CI2612 floorplan before harden.

6.3 FPGA bring-up boot - follows Kaushik's SoC4

Booting on FPGA is the same as Kaushik's SoC4 version. The Arty A7-100T bring-up boot sequence is not re-invented for DG32: it follows the SoC4 boot flow used on Kaushik's build, unchanged. Same frequencies as silicon (50 MHz CPU, 100 MHz clk_fast), so the boot ROM's 115200-baud UART divider is correct without changes; the boot banner appears on the FT2232 UART, and with a QSPI NOR on the Pmod header the ROM boots the flash image exactly as on silicon.

6.4 UART debug monitor - to be added

UART debug should be added. A firmware-independent debug monitor over a UART is a planned addition, so a die with no JTAG debugger halt can still be inspected and driven during bring-up. It reuses UART1 or a third UART instance - never the boot console - and is the counterpart to the missing debugger this OpenFrame die does not have. Tracked as Open item G.

7 Communication matrix

From	To	Via	Payload / rate
Lockstep core (LSU)	All decoded slaves	single-master AXI	load/store, per transaction
Lockstep core (fetch)	Boot ROM	private AXI	instruction words, no contention
Comparator	Fault CSR	poison + cause	on store-handshake mismatch
PWM	ADC	sample_trig	1 pulse/period at the ripple null
PWM / DShot	Motor pads	pad mux	complementary PWM or DShot, boot-selected
Encoder	IRQ / CPU	edge timestamps	position + velocity
ADC	CPU	AXI 0xDB	8-bit result on poll
Supervisor	CPU	AXI 0xD7	status only; sys_rst_n not wired to reset today
Boot ROM / QSPI-XIP	fetch / CPU	private / 0x1	the entire boot path, no fallback
DMA	SRAM / attention	AXI burst	block move, idle read port
Attention engine	Memory	AXI master via CDC	K/V load once, O writeback, on clk_fast_i
CPU	Attention engine	AXI-lite via CDC	shape programming, 0xE0000000

Isolation invariant. The frozen flight-control core touches only its own fetch ROM, the decoded peripheral bus and the fault path. No peripheral or accelerator transaction can extend the core's worst-case execution time except through the bus it already masters; the attention engine lives behind a CDC and cannot stall the CPU domain.

8 Open items

	Item	Note
A	Lockstep I/O compare window	Decide the compare-window policy before any I/O-polling firmware runs with the checker enabled on silicon (Design premise 3). The core stays frozen; this is a firmware/bring-up decision.
B	Trap handler in boot ROM	Install a real handler at 0x80000000 so a genuine PMA fault is observable, not free-running at address 0.

	Item	Note
C	Supervisor rail-fault restart	Wire sys_rst_n through an edge-qualified latch so a post-arm fault can restart the die without the cold-boot deadlock.
D	Peripheral register-map review	Confirm per-block register/IRQ/CS layouts before the shuttle freeze (Section 6.1).
E	Analog corner	Confirm analog-net dont_touch and pad ring against the shuttle floorplan (Section 6.2).
F	FPGA boot = SoC4	Bring-up boot follows Kaushik's SoC4 flow unchanged; verify against that reference build during Arty bring-up (Section 6.3).
G	UART debug monitor	Add the firmware-independent UART debug path for a JTAG-less die (Section 6.4).
H	DMA / attention bus attachment	Choose arbiter vs idle SRAM read port for the 2DOM feed and re-verify.
I	CDC constraint review	Confirm the set_clock_groups / max-delay constraints on the clk_i<->clk_fast_i crossings for the CI2612 harden.

Figures are the measured post-route numbers of the DG32-2DOM A3 die (CI2612) and the shared DG32-LITE base blocks unless marked as a target. Register-map and interrupt-source details are the peripheral review of Section 5.1. The flight-control core is frozen until the silicon bring-up campaign.

9 Register map (all blocks)

Authoritative base-address decode from the top-level decoder (addr[31:28]; the 0xD page sub-decodes addr[27:24], so each 0xD peripheral base is 0xDX00_0000). Offsets below are byte offsets within each block, extracted from the RTL and cross-checked against the AXI write/read decode.

Base	Block	Note
0x1000_0000	QSPI-XIP read window	data read window (loads); NOT instruction fetch
0x9000_0000	SRAM	32 KB main data memory
0xC000_0000	DShot ESC engine	pin-muxed onto the PWM pads
0xD000_0000	UART0	console / boot banner
0xD100_0000	UART1	telemetry
0xD200_0000	SPI master	
0xD300_0000	GPIO	14 pins on the die
0xD400_0000	TIMER0	
0xD500_0000	TIMER1	
0xD600_0000	Fault CSR	poison + cause
0xD700_0000	Supervisor / wWDT	
0xD800_0000	PWM	3-phase center-aligned
0xD900_0000	I2C master	
0xDA00_0000	Encoder	quadrature + Hall
0xDB00_0000	SAR ADC	8-bit differential
0xDC00_0000	DMA	
0xDD00_0000	SYSCTL	device ID / clock gate
0xDE00_0000	CORDIC	Q1.31
0xDF00_0000	IRQ aggregator	
0xE000_0000	INT8 attention	DG32-2DOM only (INCLUDE_ATTEN)

UART0 / UART1 (0xD0 / 0xD1)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [31:16]=clkdiv (baud = clk_i/clkdiv)
0x04	TX	W	[7:0]=byte to send (8N1); write back-pressures while tx_busy - never drops a byte
0x08	RX	R	[7:0]=received byte; read clears rx_valid
0x0C	STATUS	R	[0]=tx_busy; [1]=rx_valid

8N1: 1 start + 8 data LSB-first + 1 stop. RX 2-flop synchronised. 115200 baud at 50 MHz = 434 clocks/bit.

SPI master (0xD2)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=cpol; [2]=cpha; [9:4]=xfer_bits (8/16/32); [31:16]=clkdiv
0x04	TX	W/R	write launches a full-duplex MSB-first transfer when enabled and not busy
0x08	RX	R	last received word
0x0C	STATUS	R	[0]=busy
0x10	CS	R/W	[0]=cs_n level (reset 1 = deasserted)

SCLK idle level = CPOL. Transfer = 2 x xfer_bits SCLK edges. Hardware master, 25 MHz, 0.64 us per 16-bit word.

GPIO (0xD3)

Offset	Name	R/W	Bits / meaning
0x00	DATA_OUT	R/W	[N-1:0] value driven on pins whose OE bit is set
0x04	OE	R/W	[N-1:0] 1=drive, 0=high-Z input (reset 0 = all inputs); active-high
0x08	DATA_IN	R	[N-1:0] 2-flop synchronised pin state; writes accepted and discarded
0x0C	SET	W1S/R	write 1 sets that DATA_OUT bit; reads as DATA_OUT
0x10	CLR	W1C/R	write 1 clears that DATA_OUT bit; reads as DATA_OUT

SET/CLR are separate addresses onto one DATA_OUT register (atomic single-pin toggling). N=14 on the die.

TIMER0 / TIMER1 (0xD4 / 0xD5)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=auto_reload; [2]=irq_enable; write [8]=clear pending (W1C)
0x04	RELOAD	R/W	period-1 (fires every RELOAD+1 clocks)
0x08	COUNT	R	current down-counter value
0x0C	STATUS	R	[0]=pending (cleared via CTRL[8])

Down-counter; pulses at 0; irq = pending and irq_enable. Enable-rising reloads count.

Fault CSR (0xD6)

Offset	Name	R/W	Bits / meaning
0x0	STATUS	R	[0]=poison; [3:1]=cause. Sticky; cleared only by reset
0x4	INJECT	W	write MAGIC 0xDEAD_5AFE -> corrupt pulses 32 cycles. Reads 0
0x8	EVENTS	R	matched lockstep-compare count
0xC	ID	R	0xD632_0001

cause 001 = DATA_MISMATCH. Sticky-first: holds the first cause across continued execution.

Supervisor / wWDT (0xD7)

Offset	Name	R/W	Bits / meaning
0x00	ID	R	0x00D6_0006
0x04	CTRL	R/W	[0]=global_en; [1]=wdt_en; [7:4]=rail_en[3:0]
0x08	DEGLITCH	R/W	[7:0] deglitch cycles (reset 8)
0x0C	HOLDOFF	R/W	[15:0] reset-release holdoff (reset 1000)
0x10	WDTMIN	R/W	[15:0] watchdog window min (reset 100)
0x14	WDTMAX	R/W	[15:0] watchdog window max (reset 50000)
0x18	KICK	W	write 0xA5A5 = valid watchdog kick
0x1C	STATUS	R/W1C	[0]=sys_rst_n; [1]=wdt_fault; [7:4]=rail_sticky; [15:8]=fault_cnt

Windowed watchdog: a kick before WDTMIN or after WDTMAX both fault. Arms only when firmware sets global_en.

PWM (3-phase) (0xD8)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=EN
0x04	PERIOD	R/W	counter top, 0..65535 (period = 2*(PERIOD+1) cycles, center-aligned)
0x08	DEADTIME	R/W	dead-time cycles, all 3 legs (8-bit, max 255 = 5.1 us at 50 MHz)
0x0C	DUTY_A	R/W	phase A shadow duty, 0..PERIOD
0x10	DUTY_B	R/W	phase B shadow duty
0x14	DUTY_C	R/W	phase C shadow duty
0x18	STATUS	R	[0]=BRAKED (sticky)
0x1C	BRAKE_CLR	W	write [0]=1 to clear BRAKED once brake_i is low again

Shadow duties latch at cnt==0; sample_trig pulses there. Async hardware brake forces all 6 outputs off in <=2 cycles.

I2C master (0xD9)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [31:16]=clkdiv (SCL = clk_i/(4*(clkdiv+1)); reset 124 -> 100 kHz)
0x04	CMD	W/R	[7:1]=7-bit addr; [0]=read/write; [15:8]=byte count 1..4; [16]=send STOP
0x08	TXD	R/W	up to 4 bytes to send, LSB first on the wire
0x0C	RXD	R	up to 4 bytes received, LSB first
0x10	STATUS	R/W1C	[0]=busy; [1]=nack_addr; [2]=nack_data; [3]=arb_lost (write 1 to clear)

7-bit, 100/400 kHz, clock stretching, open-drain (external pull-ups). 2-byte read ~72 us at 400 kHz.

Encoder (quad + Hall) (0xDA)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=EN (gates quad decode + Hall capture)
0x04	POSITION	R/W	signed 4x quadrature position; firmware may write to preset/zero
0x08	INDEX_POSITION	R	POSITION latched at last Z rising edge
0x0C	STATUS	R/W1C	[0]=INDEX_SEEN (sticky); [1]=DIR (live)
0x10	TIMESTAMP	R	free-running 32-bit cycle counter
0x14	HALL_STATE	R	[2:0]=synchronised {hall_c, hall_b, hall_a}
0x18	HALL_TIMESTAMP	R	TIMESTAMP at last Hall transition
0x1C	HALL_EVENT	R/W1C	[0]=NEW_EVENT (sticky)

Edge timestamps give velocity from time-between-edges, accurate at low speed.

SAR ADC (0xDB)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=cal; [31:16]=clkdiv (reset 24 -> 50 MHz/25 = 2 MHz)
0x04	START	W	any write starts one conversion when enabled and not busy
0x08	RESULT	R	[7:0]=last conversion; [8]=valid (cleared on START)
0x0C	STATUS	R	[0]=busy; [1]=valid; [2]=comp (raw comparator)
0x10	TRIM	R/W	[4:0]=trim bits (reset 01111). NOTE: no effect on silicon - see block note

8-bit differential SAR hard macro (1.8 V, on vccd1). ~200 kSa/s at the reset divider. The macro's trim/clkc/comp taps are walled inside its own metal and are left unrouted (routing them makes the die unroutable), so TRIM and CLKC have no effect on this silicon and STATUS.comp reads 0 - the ADC runs untrimmed. The register fields are kept for API stability.

DShot ESC engine (0xC000_0000)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	W	[0]=START (self-clearing; ignored while BUSY)
0x04	BITPERIOD	R/W	cycles per bit slot (12-bit)
0x08	T0H	R/W	high-time for logical 0
0x0C	T1H	R/W	high-time for logical 1
0x10	THROTTLE0	R/W	[10:0]=throttle; [11]=telemetry request; channel 0
0x14	THROTTLE1	R/W	channel 1
0x18	THROTTLE2	R/W	channel 2
0x1C	THROTTLE3	R/W	channel 3
0x20	STATUS	R	[0]=BUSY

16-bit frame MSB-first: [15:5] throttle, [4] telemetry, [3:0] CRC4. All 4 channels share one bit-time counter.

CORDIC (0xDE)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=START (self-clearing); [1]=MODE (0=rotation, 1=vectoring)
0x04	XIN	R/W	Q1.31 x input
0x08	YIN	R/W	Q1.31 y input
0x0C	ZIN	R/W	Q1.31 angle/pi
0x10	XOUT	R	Q1.31 result x (rotation: cos; vectoring: magnitude/K)
0x14	YOUT	R	Q1.31 result y (rotation: sin)
0x18	ZOUT	R	Q1.31 result z (vectoring: atan2/pi)
0x1C	STATUS	R	[0]=BUSY

N=20 iterations, ~53-58 cycles/op. K=0x4DBA76D4; vectoring magnitude is not gain-compensated by design.

IRQ aggregator (0xDF)

Offset	Name	R/W	Bits / meaning
0x00	PENDING	R/W1C	[15:0] pending; write 1 to clear
0x04	ENABLE	R/W	[15:0] mask; external int = OR(PENDING and ENABLE)
0x08	MSIP	R/W	[0] -> machine software interrupt
0x0C	RAW	R	[15:0] source levels before edge detection
0x10	SWSET	W	write 1 sets PENDING[7]
0x14	ID	R	0x00D6_1249

Sources: 0 TMR0, 1 TMR1, 2 PWM trig, 3 ADC done, 4 DMA done, 5 I2C done, 6 FAULT, 7 SW, 8 ATTN (2DOM).

DMA (0xDC)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=start (self-clearing); [2]=src_incr; [3]=dst_incr
0x04	SRC	R/W	source address, word-aligned
0x08	DST	R/W	destination address, word-aligned
0x0C	LEN	R/W	transfer length in words (0 = none)
0x10	STATUS	R/W1C	[0]=busy; [1]=done (write 1 to clear); [2]=error
0x14	COUNT	R	words remaining

Single-beat AXI master; the top arbitrates LSU-priority over DMA. ~7.1 cycles/word.

INT8 attention engine (0xE000_0000) - writes

Offset	Name	R/W	Bits / meaning
0x00	CTRL	W	[0]=start (self-clearing); [1]=SKIP_LOAD (reuse EXP/K/V from previous kick)
0x04	A_BASE	W	Q base byte address
0x08	A_PITCH	W	Q position pitch (bytes)
0x0C	B_BASE	W	K base byte address
0x10	B_PITCH	W	K position pitch (bytes)
0x14	D_BASE	W	O (output) base byte address
0x18	D_PITCH	W	O position pitch (bytes)
0x1C	M	W	MROWS: query rows this kick (1..65535)
0x20	N	W	NK: keys (1..MAXK=400)
0x24	K	W	KD: Q/K head dim in bytes (mult of LANES and 4)
0x28	SHIFT	W	EXP index right-shift (6-bit)
0x2C	C_BASE	W	V base byte address
0x30	C_PITCH	W	V position pitch (bytes)
0x34	DV	W	output channels/head in bytes (mult of LANES and 4)
0x38	E_BASE	W	EXP table base (256 u15 entries)
0x3C	RATIO	W	u32 16.16 window constant; $rq = \text{floor}((\text{RATIO} \ll 16) / Z)$

INT8 attention engine (0xE000_0000) - reads

Offset	Name	R/W	Bits / meaning
0x00	STATUS	R	[0]=busy; [1]=done; [2]=Z==0 seen; [3]=rq overflow; [4]=EXP clamp; [5]=geometry rejected
0x40	Z	R	Z of the last completed row
0x44	rq	R	rq of the last completed row

Params: LANES=16, MAXK=400, KD<=32, DV<=64, EXP_N=256, num seat 40-bit, USE_DSP=no.

10 Block area breakdown (measured)

Std-cell logic is measured per block by yosys synthesis against the sky130A sky130_fd_sc_hd tt_025C_1v80 liberty (synth -flatten, dfflibmap, abc; chip area in mm2). Hard-macro area is the LEF footprint. The boot ROM is baked std-cell and content-dependent (measured ~0.001 mm2 with the sparse boot image). The per-block logic sum matches the measured top-level synth area (0.537 mm2, results/top_metrics.json), which validates the method.

10.1 Std-cell logic blocks (ascending)

Block	Area (mm2)	x	Subtotal (mm2)
Pad reset glue (dgrid_pad_glue)	0.0000	1	0.0000
Fault CSR (dgrid_fault_csr)	0.0019	1	0.0019
JTAG TAP (dgrid_jtag_tap)	0.0022	1	0.0022
IRQ controller (dgrid_irq)	0.0035	1	0.0035
Sysctl (dgrid_sysctl)	0.0036	1	0.0036
ADC digital wrapper (dgrid_adc)	0.0037	1	0.0037
GPIO (dgrid_gpio)	0.0039	1	0.0039
CDC lite bridge (dgrid_axi_cdc_lite)	0.0051	1	0.0051
Timer (dgrid_timer)	0.0059	2	0.0117
UART (dgrid_uart)	0.0067	2	0.0134
Supervisor + wWDt (dg_supervisor)	0.0077	1	0.0077
I2C master (dgrid_i2c)	0.0085	1	0.0085
Encoder (dgrid_encoder)	0.0092	1	0.0092
SPI master (dgrid_spi)	0.0093	1	0.0093
DShot engine (dgrid_dshot)	0.0101	1	0.0101
PWM 3-phase (dgrid_pwm)	0.0113	1	0.0113
DMA (dgrid_dma)	0.0123	1	0.0123
SRAM controller logic (dgrid_axi_sram)	0.0156	1	0.0156
CORDIC (dgrid_cordic)	0.0208	1	0.0208
Lockstep comparator (dgrid_lockstep_check)	0.0397	1	0.0397
INT8 attention engine logic (dgrid_int8_attn)	0.0694	1	0.0694
CDC burst-read bridge (dgrid_axi_cdc_burst_rd)	0.0745	1	0.0745
CDC burst-write bridge (dgrid_axi_cdc_burst_wr)	0.0835	1	0.0835
CPU core, each (VexiiRiscv rv32imc)	0.1648	2	0.3297

Logic subtotal: **0.751 mm2**.

10.2 Hard macros

Block	Area (mm2)	x	Subtotal (mm2)
Boot ROM (baked, sparse image)	0.0010	1	0.0010
SAR ADC macro	0.0540	1	0.0540
Main SRAM macro (sky130 2kB 1rw1r)	0.2845	16	4.5527
Attention K buffer SRAM macro	0.2845	1	0.2845
Attention V buffer SRAM macro	0.2845	1	0.2845

Macro subtotal: **5.177 mm2**.

Memory macros are the die; logic is small. The 18 SRAM macros are 5.12 mm2 - about 86% of the placed cell+macro area (5.93 mm2); every std-cell logic block together, including both lockstep CPU cores, is only 0.75 mm2. The die is 15.30 mm2, so the rest is PDN, placement halos and routing channels. The checker is a mirror core (CHK_MIRROR, DCLS): it has NO private memory, so there is no checker RAM to remove. The only sub-mm2 lever that costs neither throughput nor the lockstep pair is main-SRAM capacity - dropping 2 of the 16 main macros takes 32 KB to 28 KB and saves 0.57 mm2, if the firmware image and data fit.

11 Application: AVIP bearing-fault detection

The DG32-2DOM attention engine exists to run models like **AVIP**, an INT8 bearing-fault classifier (dgrid-a8w8). AVIP is multimodal - a time branch and an FFT branch, each a small conv stack plus a gated recurrent cell - fused by two self-attention blocks and one cross-attention block, ending in a per-class head. Those attention blocks are exactly what the engine accelerates. The shipped CWRU-trained model is 47,076 parameters at 94-95% held-out-load accuracy with INT8 equal to FP32; the full AVIP topology is 478,277 parameters. Both map onto the same op set with no new RTL.

11.1 What runs on the attention engine

Model stage	What it is	DG32-2DOM mapping
Self-attention (sa, sb)	QKV projection + softmax(QK ^T)V per head	K_ATTN -> dgrid_int8_attn
Cross-attention (cx)	time<->freq fusion, one head set	K_ATTN -> dgrid_int8_attn
Head dimension	key_dim 16 -> KD=16	multiple of LANES=16, within KDMAX=32 - fits as built
Kicks per inference	6 (CWRU model) / 12 (full AVIP)	one kick per (window, head)
Conv / elementwise / dense	the rest of the graph	K_CONV / K_ELEM / K_AGEMM on the INT8 tile

11.2 Inference budget (measured on the INT8 op-graph)

Model	Positions	Kicks	Conv cycles	Time (ms)
CWRU model (47K params)	143,377	228,229	3,794,371	46.7
Full AVIP (478K params)	169,821	231,657	4,429,355	54.5

AVIP has 10x the parameters but only 1.2x the compute: its capacity is in deeper, narrower layers at lower spatial resolution, and compute tracks positions, not parameter count. The engine runs on clk_fast_i, so the ~50 ms inference does not steal cycles from the 50 MHz control core.

Footprint. The AVIP arena is ~2.49 MB (weights, LUTs, activations), so the full model runs from external memory on the DG32-2DOM / a8w8 class part - not from the 32 KB on-chip SRAM. The on-chip attention engine and INT8 tile are shared; the model image and activations live off-chip.

11.3 Current-signature front end (no accelerometer)

AVIP trains on two accelerometers (drive-end + fan-end). A motor-control die already has the sensor it needs: the phase-current SAR ADC. A bearing defect modulates the stator current at $f_e \pm k \cdot f_{\text{fault}}$ (Current Signature Analysis), so the current envelope carries the same fault signature. The front end (fw/avip_csa) samples the SAR ADC at the PWM ripple null, derives two channels from the one current (raw carrier + AM envelope), and produces both a standalone classical fault score and the AVIP INT8 recurrence tensor.

Item	How	Note
Sensor	On-die SAR ADC (0xDB), PWM-triggered	no external accelerometer or vibration probe
Channels	current + envelope	the DE/FE analogue from a single ADC
Classical indicator	BPFO/BPFI/BSF sideband energy, 0..255	~9 KB, fits DG32-LITE; runs every window
NN input	(35,20,20,2) INT8 recurrence tensor	streamed to the arena for the op-graph on 2DOM

The classical indicator is a usable detector on its own and needs no accelerator, so it also runs on DG32-LITE; the trained-network path adds accuracy on DG32-2DOM. Bearing-fault detection is therefore a concrete reason the attention variant exists, and it reuses the motor's own current sense.